

Mathematics Bootcamp

Optimization and Economic Choice

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Today

- Existence
 - Maximum versus maximizers; when a solution is attained
- Candidates and Verification
 - First- and second-order conditions; concavity
- Constraints
 - Equality constraints, Lagrange multipliers, and a brief look at Kuhn–Tucker
- Economic Applications
 - Consumer demand and static firm choice

Optimization Has Two Separate Questions

Consider the scalar problem

$$\max\{f(x) : x \in D\}, \quad D \subseteq \mathbb{R}.$$

Existence

Does some feasible point actually attain the largest value?

This is a question about the feasible set and the objective together.

Identification

If a maximizer exists, what conditions must it satisfy?

Derivatives generate candidates; they do not create a solution.

Day 1 wrote choice as an *argmax*. Today we study what makes that expression meaningful.

The Maximum and the Maximizers Are Different Objects

Maximum Value

If the largest value is attained, then

$$\max\{f(x) : x \in D\} = f(x^*).$$

The maximum is a number.

Maximizers

$$\arg \max_{x \in D} f(x)$$

is the set of feasible x such that $f(x) \geq f(y)$ for every $y \in D$.

The maximum records how high the objective goes; the argmax records where it gets there.

The Argmax Can Be Empty, a Singleton, or Many Points

Problem	Argmax	What happens
$\max_{x \in (0,1)} x$	$\emptyset$	the supremum is not attained
$\max_{x \in [-1,1]} (-x^2)$	$\{0\}$	one maximizer
$\max_{x \in [0,1]} 0$	$[0, 1]$	many maximizers

Existence and Uniqueness Are Separate

An empty argmax means no maximizer exists. A nonempty argmax may contain one point or many points.

Continuity and Compactness Guarantee Extrema

Extreme Value Theorem: Practical Form

If the choice set $D \subseteq \mathbb{R}$ is nonempty and compact and the objective $f : D \rightarrow \mathbb{R}$ is continuous, then there are points $x^{\max}, x^{\min} \in D$ such that

$$f(x^{\max}) \geq f(x) \geq f(x^{\min}) \quad \text{for every } x \in D.$$

In One Dimension

Compact means closed and bounded. Thus a closed bounded interval $[a, b]$ is the canonical compact choice set.

Continuous objective + compact choice set $\implies$ maximum and minimum are attained.

These conditions guarantee existence, not uniqueness.

Unconstrained Means Interior to the Feasible Set

Local Maximum

x^* is a local maximizer if there is some $\varepsilon > 0$ such that

$$f(x^*) \geq f(x)$$

whenever $x \in D$ and $|x - x^*| < \varepsilon$.

Global Maximum

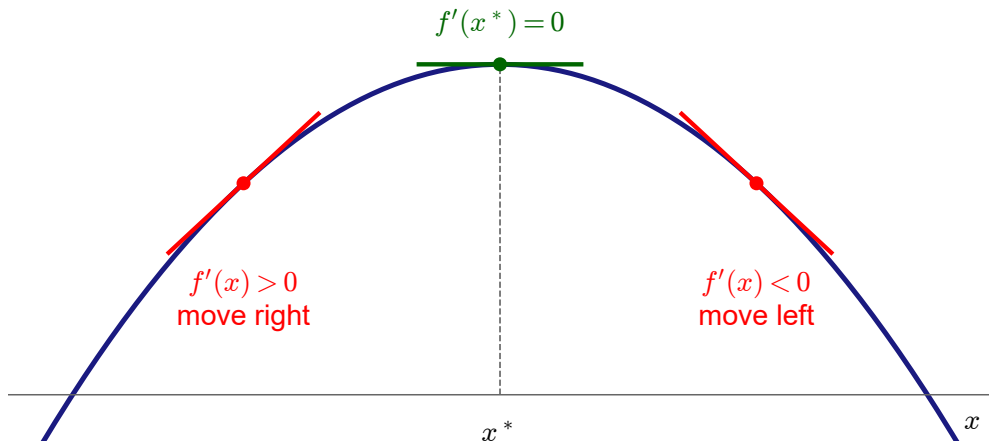
x^* is a global maximizer if

$$f(x^*) \geq f(x)$$

for every $x \in D$.

An *unconstrained* first-order calculation assumes that x^* is an interior point: sufficiently small movements both left and right are feasible.

A Nonzero Slope Leaves an Improving Direction



If $f'(x^*) > 0$, moving slightly right raises f ; if $f'(x^*) < 0$, moving slightly left raises f .

Why the First-Order Condition Is Necessary

First-Order Necessary Condition

If x^* is an interior local maximizer and f is differentiable at x^* , then $f'(x^*) = 0$.

- (1) For sufficiently small $h > 0$, local optimality gives

$$\frac{f(x^* + h) - f(x^*)}{h} \leq 0.$$

- (2) For sufficiently small $h < 0$, the numerator is still nonpositive but the denominator is negative, so

$$\frac{f(x^* + h) - f(x^*)}{h} \geq 0.$$

- (3) Differentiability makes the two one-sided limits equal. Thus $f'(x^*) \leq 0$ and $f'(x^*) \geq 0$, which implies $f'(x^*) = 0$.

A Critical Point Is Only a Candidate

A point satisfying $f'(x) = 0$ is a *critical point*. The condition alone does not identify its role.

Function	At $x = 0$	Conclusion
$f(x) = -x^2$	$f'(0) = 0$	strict maximum
$f(x) = x^2$	$f'(0) = 0$	strict minimum
$f(x) = x^3$	$f'(0) = 0$	neither

Boundary Warning

A boundary maximizer need not satisfy $f'(x) = 0$, because movement may be feasible in only one direction.

The Second Derivative Gives a Local Test

Suppose f is twice continuously differentiable on a neighborhood of x^* .

Necessary at a Smooth Maximum

If x^* is an interior local maximizer and $f''(x^*)$ exists, then

$$f''(x^*) \leq 0.$$

Sufficient for a Strict Local Maximum

If

$$f'(x^*) = 0 \quad \text{and} \quad f''(x^*) < 0,$$

then x^* is a strict local maximizer.

If $f''(x^*) = 0$, the test is inconclusive:

x^4 has a minimum, $-x^4$ has a maximum, x^3 has neither at 0.

Concavity Turns a Local Condition into a Global Conclusion

Concavity on an Interval

For every x, y in the interval and every $t \in [0, 1]$,

$$f(tx + (1 - t)y) \geq tf(x) + (1 - t)f(y).$$

Global First-Order Test

If f is differentiable and concave on an interval, then

$$f'(x^*) = 0 \implies x^* \text{ is a global maximizer.}$$

If f is strictly concave, there is at most one maximizer.

For a twice differentiable scalar function, $f''(x) \leq 0$ throughout an interval is a convenient test for concavity.

Existence, Candidates, and Comparison Do Different Jobs

For a continuous function on a closed bounded interval $[a, b]$:

1. **Existence:** Weierstrass guarantees that a maximizer exists.
2. **Interior candidates:** solve $f'(x) = 0$ for $x \in (a, b)$.
3. **Boundary candidates:** include a and b .
4. **Verification:** compare the objective at every candidate.

Why First-Order Conditions Are Not Enough

They narrow the search. They neither guarantee that a maximizer exists nor establish that a candidate is globally best.

An Equality Constraint Leaves One Free Variable

Consider

$$\max_{x,y} f(x, y) \quad \text{subject to} \quad g(x, y) = 0.$$

Suppose that, near a candidate (x^*, y^*) , the constraint determines $y = q(x)$. Then the constrained problem becomes

$$\max_x F(x), \quad F(x) = f(x, q(x)).$$

Differentiating the constraint $g(x, q(x)) = 0$ gives

$$g_x(x, q(x)) + g_y(x, q(x))q'(x) = 0, \quad q'(x) = -\frac{g_x}{g_y},$$

provided $g_y \neq 0$. Here $g_x = \partial g / \partial x$ and $g_y = \partial g / \partial y$.

The Chain Rule Produces the Tangency Condition

At an interior constrained optimum,

$$\begin{aligned} 0 = F'(x^*) &= f_x + f_y q'(x^*) \\ &= f_x - f_y \frac{g_x}{g_y}. \end{aligned}$$

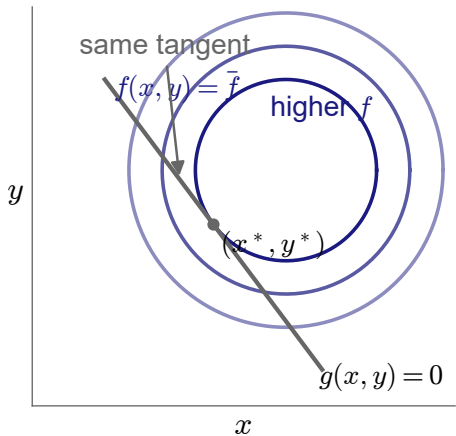
Therefore,

$$f_x g_y = f_y g_x.$$

When the relevant derivatives are nonzero,

$$\boxed{-\frac{f_x}{f_y} = -\frac{g_x}{g_y}.$$

The objective level curve and the constraint have the same slope.



The Constraint Selects a Feasible Curve

Consider

$$f(x_1, x_2) = 4 - (x_1 - 1)^2 - (x_2 - 1)^2$$

subject to

$$x_1 + x_2 = 1.6.$$

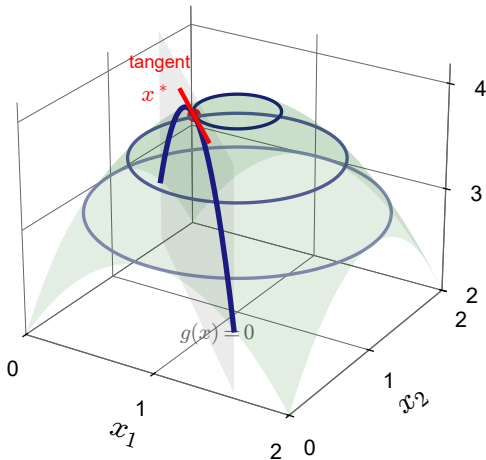
The constraint cuts the blue feasible curve from the green objective surface. Along that curve,

$$\phi(x_1) = f(x_1, 1.6 - x_1).$$

Its highest point is

$$(x_1^*, x_2^*) = (0.8, 0.8),$$

$$\phi'(0.8) = 0.$$



The Lagrangean Packages the Same Calculation

Define the Lagrangean

$$L(x, y; \lambda) = f(x, y) + \lambda g(x, y).$$

Stationarity

$$\frac{\partial L}{\partial x} = f_x + \lambda g_x = 0,$$

$$\frac{\partial L}{\partial y} = f_y + \lambda g_y = 0.$$

Eliminating λ gives $f_x g_y = f_y g_x$.

These equations generate candidates; they do not prove global optimality.

Feasibility

$$\frac{\partial L}{\partial \lambda} = g(x, y) = 0.$$

The candidate must remain on the constraint.

Regularity Makes the Multiplier Exist and Be Unique

Theorem of Lagrange: One Equality

Let U be an open neighborhood of (x^*, y^*) , and suppose f and g are continuously differentiable on U . If (x^*, y^*) is a local optimum on $U \cap \{(x, y) : g(x, y) = 0\}$, and at least one of $g_x(x^*, y^*)$ and $g_y(x^*, y^*)$ is nonzero, then there is a multiplier λ^* satisfying

$$f_x(x^*, y^*) + \lambda^* g_x(x^*, y^*) = 0, \quad f_y(x^*, y^*) + \lambda^* g_y(x^*, y^*) = 0.$$

The same regularity condition makes λ^* unique *at the fixed optimizer* (x^*, y^*) .

What the Theorem Does Not Say

It does not establish that an optimizer exists, and it does not make the optimizer unique.

The Lagrange Procedure Starts Before the Lagrangean

1. **Establish existence.** Show that the original constrained problem has a global optimum.
2. **Check regularity.** Verify that g_x and g_y are not both zero at feasible candidates.
3. **Generate candidates.** Solve $L_x = 0$, $L_y = 0$, and $L_\lambda = 0$.
4. **Verify.** Compare feasible candidates, or use concavity to establish global optimality.

Two Ways the Cookbook Can Fail

If no optimum exists, there is nothing to characterize. If regularity fails, an optimum may exist without any multiplier solving the stationarity equations.

A Multiplier Can Measure the Value of a Constraint Change

Let the constraint depend on a scalar parameter c :

$$g(x, y) + c = 0.$$

Suppose the optimizer $(x(c), y(c))$ and its multiplier $\lambda(c)$ vary smoothly, and define the optimized value

$$V(c) = f(x(c), y(c)).$$

With the sign convention $L = f + \lambda[g + c]$,

$$\boxed{V'(c) = \lambda(c).}$$

- $\lambda(c)$ is the marginal change in optimized value from increasing c .
- This interpretation needs the additional smoothness just stated.
- The number changes if the objective or the constraint is rescaled.

Kuhn–Tucker Adds a Boundary Condition

Consider the one-dimensional inequality problem

$$\max_{x \geq 0} f(x), \quad L(x; \lambda) = f(x) + \lambda x.$$

Interior Candidate

If $x^* > 0$, both directions are feasible:

$$f'(x^*) = 0, \quad \lambda^* = 0.$$

Boundary Candidate

If $x^* = 0$, feasible moves go only right:

$$f'(0) \leq 0, \quad \lambda^* = -f'(0) \geq 0.$$

Both cases satisfy

$$f'(x^*) + \lambda^* = 0, \quad x^* \geq 0, \quad \lambda^* \geq 0, \quad \lambda^* x^* = 0.$$

For a concave f , these conditions are also sufficient for global optimality.

Time for Economics

- The objective f becomes utility or profit.
- The feasible interval becomes a budget set or a technological restriction.
- Equality constraints describe choices on a binding budget.
- Nonnegativity creates corners, where ordinary tangency is inappropriate.

The Order of Reasoning Does Not Change

Establish existence, generate all relevant candidates, and then verify which candidate is globally best.

Consumer Demand Is an Argmax Problem

A consumer chooses two nonnegative quantities x_1 and x_2 , has income $I > 0$, and faces prices $p_1, p_2 > 0$.

Budget Set

$$B(p_1, p_2, I) = \{(x_1, x_2) : x_1, x_2 \geq 0, p_1 x_1 + p_2 x_2 \leq I\}.$$

Consumer Problem

$$(x_1^*, x_2^*) \in \arg \max_{(x_1, x_2) \in B(p_1, p_2, I)} u(x_1, x_2).$$

The set of maximizing pairs is the demand correspondence; it becomes a function when the maximizer is unique.

The Assumptions from Day 3 Now Do Separate Jobs

Assumption	Consequence
$p_1, p_2 > 0$ and $I > 0$	$0 \leq x_1 \leq I/p_1$ and $0 \leq x_2 \leq I/p_2$; the budget set is closed and bounded.
Continuity of u	At least one affordable bundle maximizes utility.
Strict quasiconcavity of u	At most one bundle maximizes utility on the convex budget set.
Local nonsatiation	Every optimal bundle exhausts the budget.

Existence, uniqueness, and budget exhaustion are three different conclusions.

Local Nonsatiation Makes the Budget Bind

Budget Exhaustion

If preferences are locally nonsatiated and (x_1^*, x_2^*) solves the consumer problem, then

$$p_1 x_1^* + p_2 x_2^* = I.$$

- (1) Suppose instead that $p_1 x_1^* + p_2 x_2^* < I$. There is positive spending slack.
- (2) Because prices are finite, every sufficiently nearby nonnegative bundle remains affordable.
- (3) Local nonsatiation gives a sufficiently nearby bundle (y_1, y_2) that is strictly preferred.
- (4) That bundle is affordable and better, contradicting optimality.

Therefore the budget inequality can be replaced by an equality when solving this problem.

A Binding Budget Reduces Consumer Choice to One Variable

Budget exhaustion gives

$$x_2 = \frac{I - p_1 x_1}{p_2}, \quad 0 \leq x_1 \leq \frac{I}{p_1},$$
$$\phi(x_1) = u\left(x_1, \frac{I - p_1 x_1}{p_2}\right).$$

At a differentiable interior optimum $x_1^* \in (0, I/p_1)$,

$$0 = \phi'(x_1^*) = u_1(x_1^*, x_2^*) - \frac{p_1}{p_2} u_2(x_1^*, x_2^*),$$

where $u_i = \partial u / \partial x_i$. Hence

$$\boxed{\frac{u_1(x_1^*, x_2^*)}{p_1} = \frac{u_2(x_1^*, x_2^*)}{p_2}}.$$

The Consumer Lagrangean Gives the Same Equations

Write

$$\begin{aligned}L(x_1, x_2; \lambda) &= u(x_1, x_2) + \lambda(I - p_1x_1 - p_2x_2), \\u_1(x_1^*, x_2^*) &= \lambda^* p_1, \\u_2(x_1^*, x_2^*) &= \lambda^* p_2, \\p_1x_1^* + p_2x_2^* &= I.\end{aligned}$$

Interior Interpretation

Each consumed good yields the same marginal utility per unit of money: $u_1/p_1 = u_2/p_2 = \lambda^*$.

This tangency condition applies only when both quantities are strictly positive and utility is differentiable.

Cobb–Douglas Has an Interior Solution

Let

$$u(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}, \quad 0 < \alpha < 1.$$

1. Utility equals zero on either axis.
2. With $I > 0$ and positive prices, an affordable bundle with $x_1, x_2 > 0$ exists and gives positive utility.
3. Therefore no optimal bundle lies on an axis: $x_1^*, x_2^* > 0$.

Thus we may first restrict attention to $x_1, x_2 > 0$. Only then may we use the strictly increasing transformation

$$\log u(x_1, x_2) = \alpha \log x_1 + (1 - \alpha) \log x_2.$$

Cobb–Douglas Demand Spends Fixed Income Shares

The first-order conditions for the transformed objective are

$$\frac{\alpha}{x_1} = \lambda p_1, \quad \frac{1 - \alpha}{x_2} = \lambda p_2, \quad p_1 x_1 + p_2 x_2 = I.$$

Multiplying the first two equations by x_1 and x_2 gives

$$\alpha = \lambda p_1 x_1, \quad 1 - \alpha = \lambda p_2 x_2.$$

Adding them and using the budget gives $\lambda = 1/I$. Therefore,

$$\boxed{x_1^* = \frac{\alpha I}{p_1}, \quad x_2^* = \frac{(1 - \alpha) I}{p_2}.$$

The consumer spends the shares α and $1 - \alpha$ of income on the two goods. The log transformation preserves the bundle, but not the numerical multiplier.

Perfect Substitutes Require a Comparison, Not a Tangency

Let

$$u(x_1, x_2) = ax_1 + bx_2, \quad a, b > 0.$$

Utility per unit of money is constant: compare a/p_1 with b/p_2 . Write $x(p_1, p_2, I)$ for the set of utility-maximizing affordable pairs.

Demand

$$x(p_1, p_2, I) = \begin{cases} \{(I/p_1, 0)\}, & a/p_1 > b/p_2, \\ \{(0, I/p_2)\}, & a/p_1 < b/p_2, \\ \{(x_1, x_2) : x_1, x_2 \geq 0, p_1x_1 + p_2x_2 = I\}, & a/p_1 = b/p_2. \end{cases}$$

The first two cases are corners; the equality case has many maximizers.

Leontief Utility Is Solved at the Kink

Let

$$u(x_1, x_2) = \min\{ax_1, bx_2\}, \quad a, b > 0.$$

Unmatched units do not raise utility, so the optimal bundle satisfies

$$ax_1 = bx_2.$$

Local nonsatiation also makes the budget bind:

$$p_1x_1 + p_2x_2 = I.$$

Solving these two scalar equations gives

$$\boxed{x_1^* = \frac{bI}{bp_1 + ap_2}, \quad x_2^* = \frac{aI}{bp_1 + ap_2}.$$

The relevant point is a kink, so ordinary derivative tangency does not apply.

A Static Firm Compares Marginal Product with the Wage

Normalize the output price to one. A one-period firm chooses labor $\ell \geq 0$:

$$\max_{\ell \geq 0} \{f(\ell) - w\ell\}.$$

Existence

A solution exists, for example, if labor is restricted to a closed bounded interval and profit is continuous.

Identification

For differentiable, concave f :

$$\ell^* > 0 \Rightarrow f'(\ell^*) = w,$$

while $f'_+(0) \leq w$ supports $\ell^* = 0$.

Here $f'_+(0)$ is the right derivative. The problem is static: hiring, production, and payment all occur in the same period.

Linear Production Reopens the Existence Question

Let one unit of labor produce $A > 0$ units of output:

$$f(\ell) = A\ell, \quad \pi(\ell) = (A - w)\ell.$$

Marginal product is constant, so there is no interior tangency to find.

Profit Maximizers without a Capacity Constraint

$$\arg \max_{\ell \geq 0} (A - w)\ell = \begin{cases} \{0\}, & A < w, \\ [0, \infty), & A = w, \\ \emptyset, & A > w. \end{cases}$$

With a capacity $0 \leq \ell \leq \bar{\ell}$, the case $A > w$ has the solution $\ell^* = \bar{\ell}$.

First-Order Conditions Narrow the Search

1. Write the objective and the feasible set.
2. Establish that a maximizer exists; ask separately whether it is unique.
3. Classify the possible solution: interior, equality-constrained, boundary, or kink.
4. Generate every candidate with derivatives, the Lagrangean, Kuhn–Tucker, or direct comparison.
5. Verify feasibility and global optimality.

The Discipline of Optimization

Existence $\longrightarrow$ candidates $\longrightarrow$ verification.